```
def cumulative_density(floor, first, last):
    return sum( ❶
        [ ❶
            marginal_probability(i, first, last) ❶
            for i in range(first, floor) ❶
        ] ❶
    ) ❶
```

❶ Sum over the probabilities for *X* less than the threshold.

Since our prior knowledge dictates that the phone won't break if dropped from the 1st floor, this probability is *0*.

4. Compute the entropy of the fictitious posterior distribution in the two cases where we conduct a trial on the 5th floor.

   Using the `compute_entropy()` function we implemented, we can compute the entropy in two cases: if the phone breaks, we set `first = 1` and `last = 4`, else we set `first = 5` and `last = 10`.

```
>>> compute_entropy(1, 4)
Output: 1.75
>>> compute_entropy(5, 10)
Output: 1.9375
```

5. Given the prior distribution, compute the expected posterior entropy after you conduct a trial on the 5th floor.

   We already have done the necessary calculations for this expected posterior entropy computation. First, the probability that the phone will break from the 5th floor is *0.9375*, in which case the posterior entropy is *1.75*. Second, the probability that the phone won't break from the 5th floor is *1 - 0.9375 = 0.0625*, in which case the posterior entropy is *1.9375*.

   Taking the average of the two cases gives: *(0.9375) 1.75 + (0.0625) 1.9375 = 1.76171875*. This is the expected posterior entropy after you conduct a trial on the 5th floor.

6. . Compute this expected posterior entropy for other floors.

   We can implement a function that does the calculation we just went through:

```
def compute_expected_posterior_entropy(floor, first, last):
    break_probability = cumulative_density(floor, first, last) ❶

    return ( ❷
        break_probability * compute_entropy(first, floor - 1) ❷
        + (1 - break_probability) * compute_entropy(floor, last) ❷
    ) ❷
```

❶ The probability that the phone will break from a given floor.

❷ Take the average of the two cases.

Using this function, we can plot out the expected posterior entropy for numbers between 1 and 10.

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